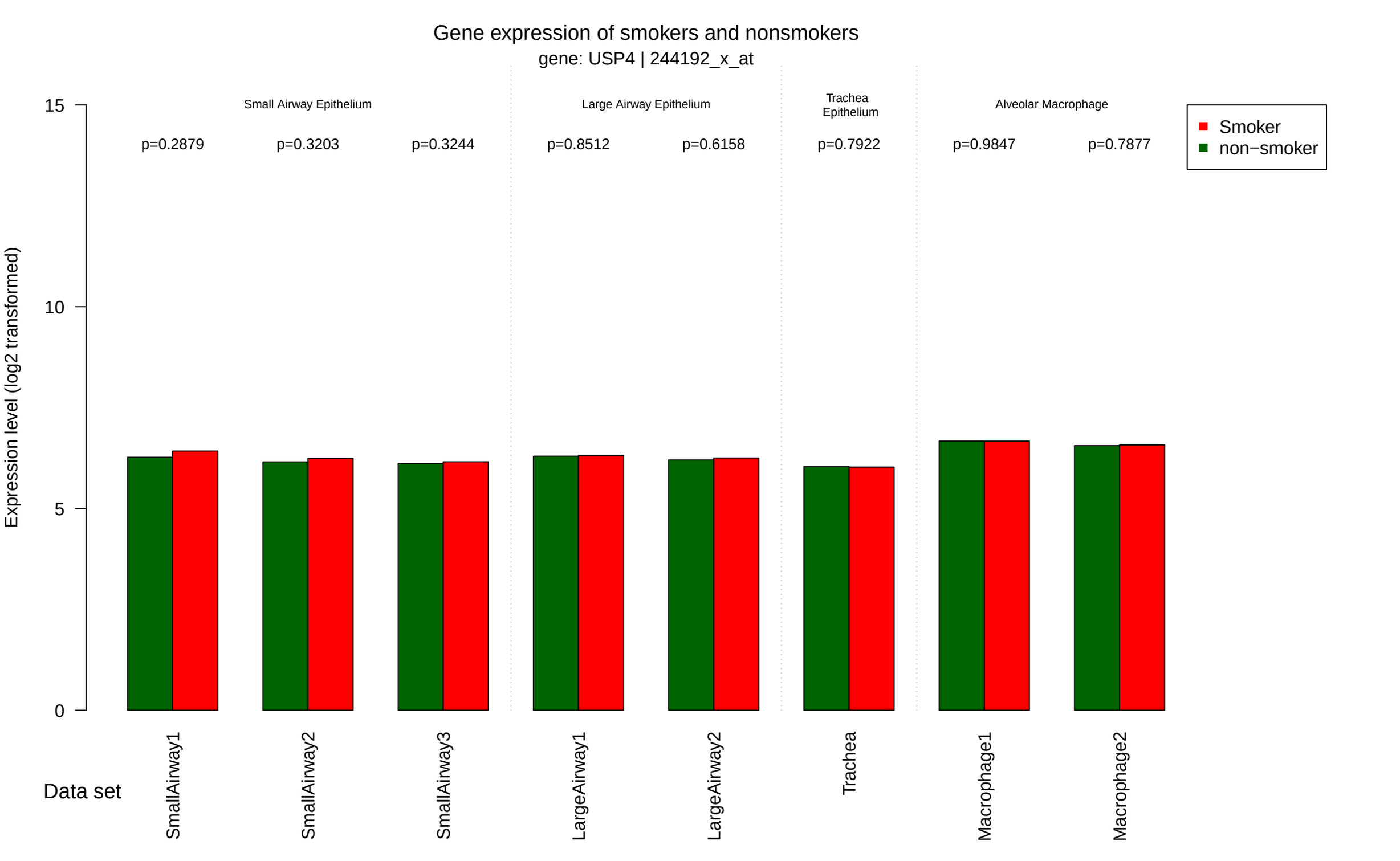
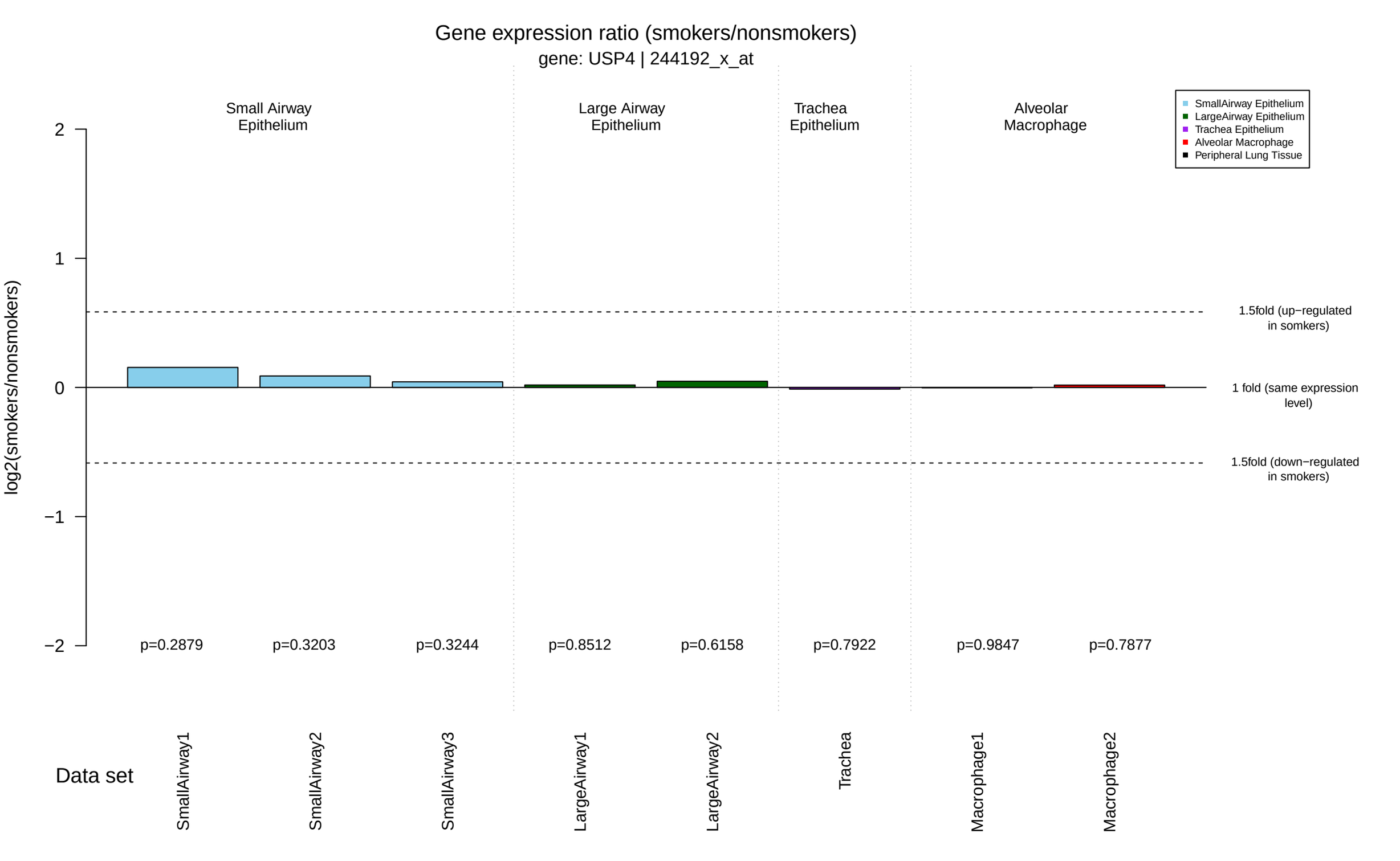
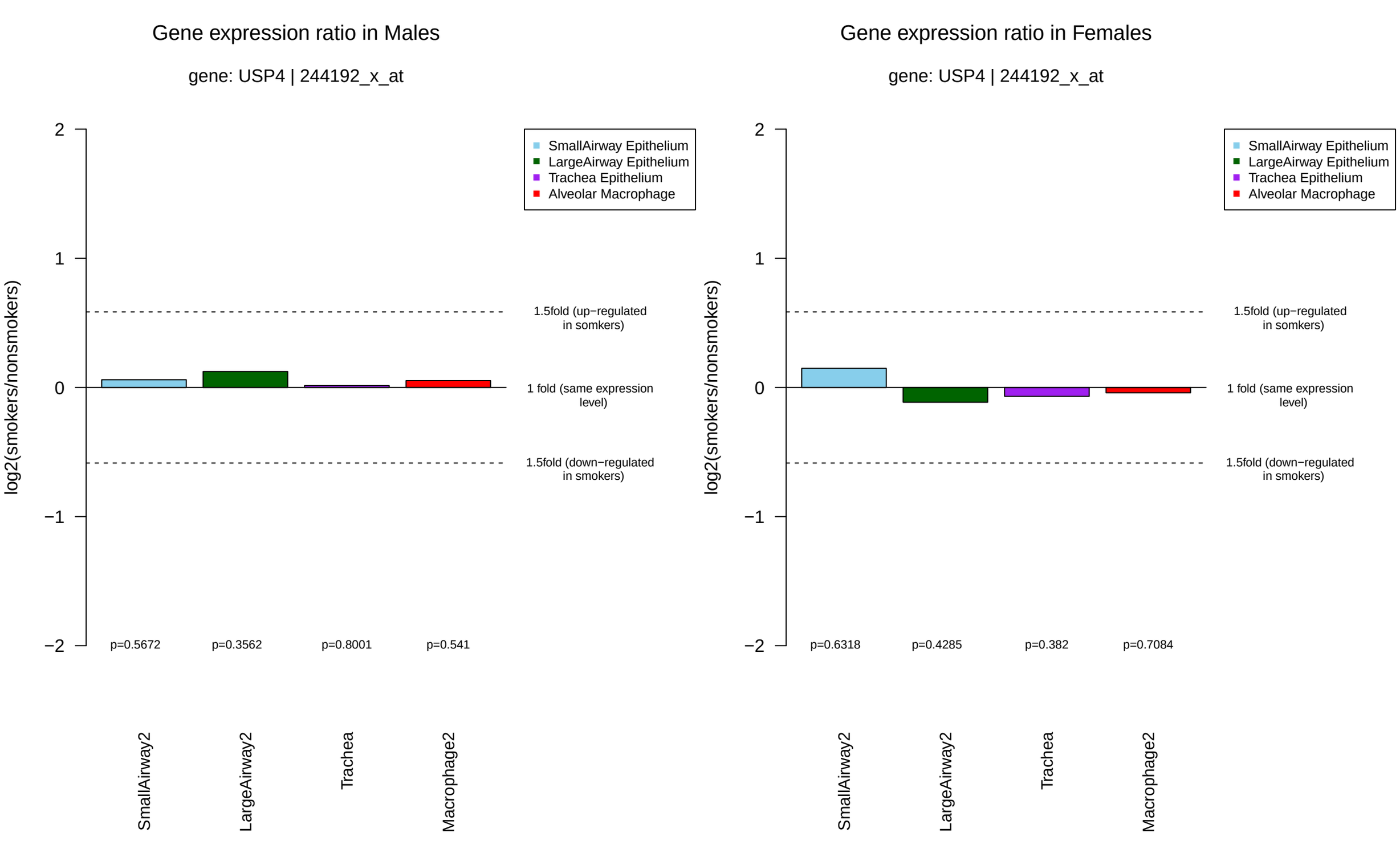
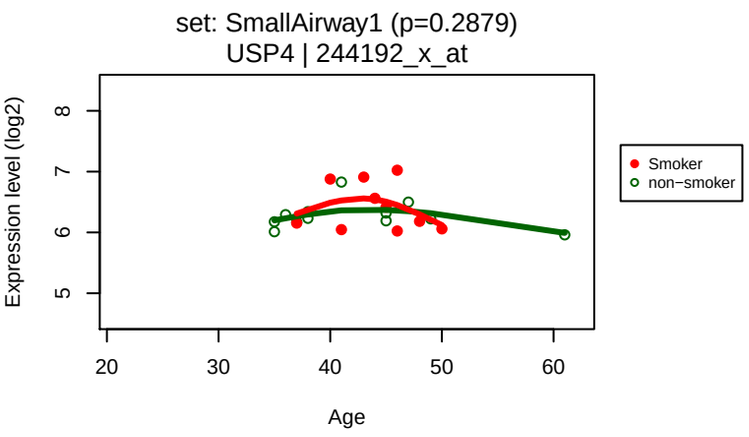
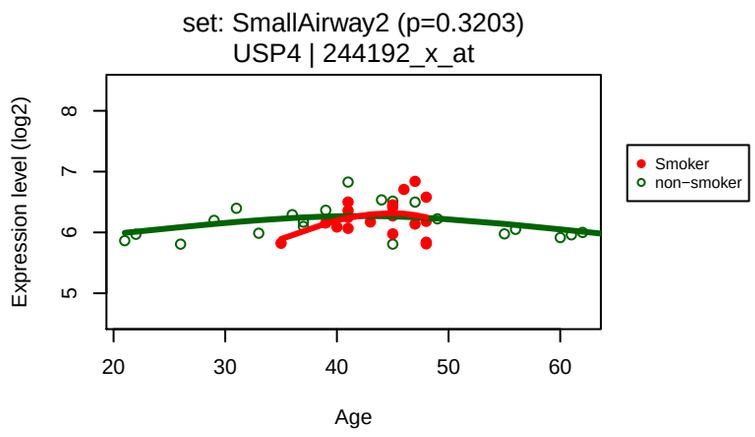
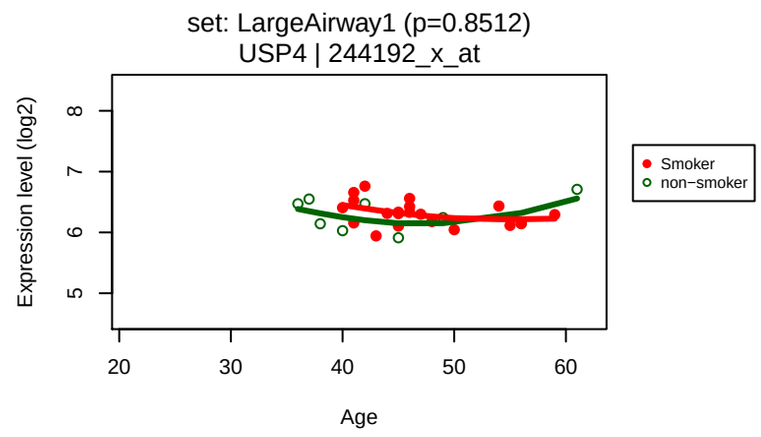

Gene expression of smokers and nonsmokers
gene: USP4 | 244192_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2879
p=0.3203
p=0.3244
p=0.8512
p=0.6158
p=0.7922
p=0.9847
p=0.7877
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

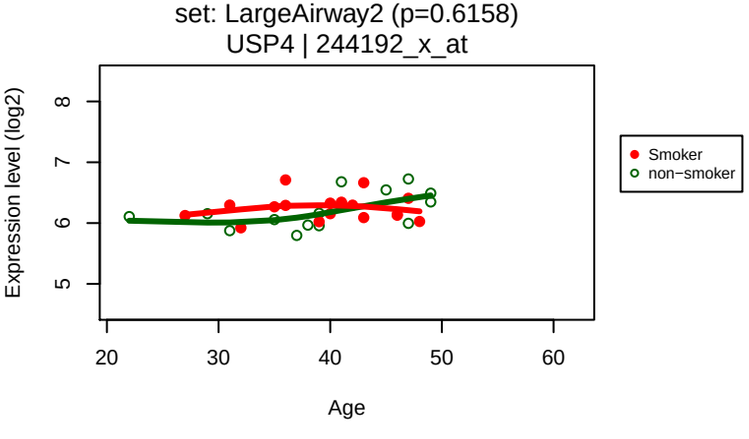

Gene expression ratio (smokers/nonsmokers)
gene: USP4 | 244192_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2879
p=0.3203
p=0.3244
p=0.8512
p=0.6158
p=0.7922
p=0.9847
p=0.7877
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

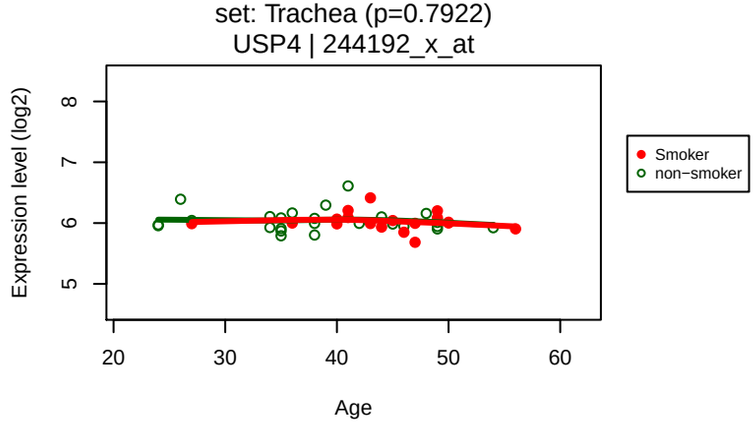

Gene expression ratio in Males
gene: USP4 | 244192_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5672
p=0.3562
p=0.8001
p=0.541
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP4 | 244192_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6318
p=0.4285
p=0.382
p=0.7084
SmallAirway2
LargeAirway2
Trachea
Macrophage2

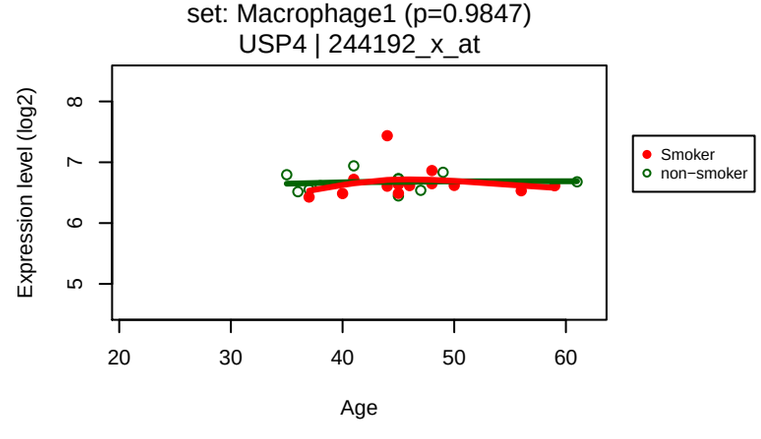

set: SmallAirway1 (p=0.2879)
USP4 | 244192_x_at
Expression level (log2)
Age
Smoker
non−smoker

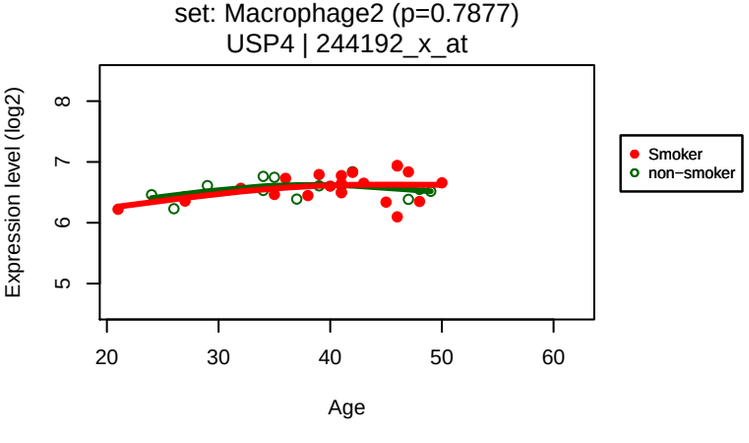

set: SmallAirway2 (p=0.3203)
USP4 | 244192_x_at
Expression level (log2)
Age
Smoker
non−smoker

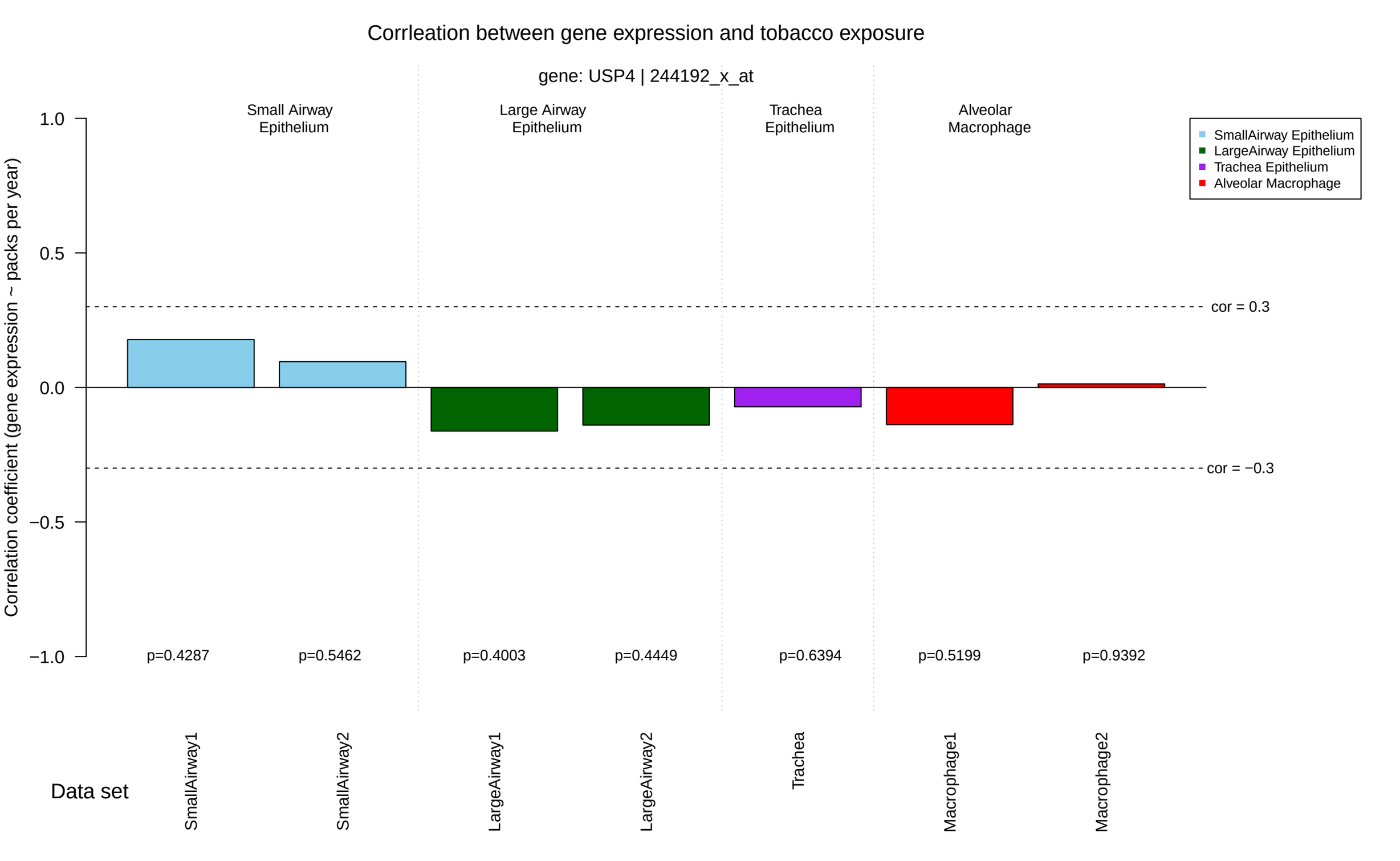

set: LargeAirway1 (p=0.8512)
USP4 | 244192_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6158)
USP4 | 244192_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7922)
USP4 | 244192_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9847)
USP4 | 244192_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7877)
USP4 | 244192_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP4 | 244192_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4287
p=0.5462
p=0.4003
p=0.4449
p=0.6394
p=0.5199
p=0.9392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2